15	**(a)**		$c = 1.14$	**B1**	**3.3**	
				[1]		
15	**(b)**		$1.20 = 4a + 2b + 1.14$ **oe** $1.25 = 16a + 4b + 1.14$ **oe**	**M1**	**3.3**	**both** equations. FT *their c*
			$a = -0.00125, b = 0.0325$	**A1**	**1.1**	Fractional equivalents are $a = -\frac{1}{800}$ and $b = \frac{13}{400}$ Equivalents in standard form is acceptable
				[2]		
15	**(c)**		$1.29 = 1.14 + 0.0325t - 0.00125t^2$	**M1**	**3.1b**	FT *their a,b,c* (Can be > etc)
			$t = 6$ and 20	**A1**	**3.4**	
			$6 \leq t \leq 20$	**A1**	**3.5a**	Set notation such as $t \in [6, 20]$ is fine but must not be soft brackets $t \geq 6$ **and** $t \leq 20$ or $t \geq 6 \cap t \leq 20$ but NOT $t \geq 6$, $t \leq 20$
				[3]		

Question			Answer	Marks	AO	Guidance
15	(d)		It will eventually predict a **negative exchange rate** oe **(will fall below zero etc)**	**B1**	**3.5a**	'Exchange rate tends to zero' is B0 Must mention the variable 'exchange rate' Underlined words needed
				[1]		